SPX Multi-Date Volatility Surface Comparison

Comprehensive Quantitative Blueprint: Mathematical Derivations, Smile Factor Decomposition, Fixed-Strike Butterfly Repricing, and Implementation Architecture

Target Asset: S&P; 500 Index (SPX)	**Reference Maturity:** ~60 DTE Target Chain
Strike Coordinate: x = (K – K_ATM) / H	**Reference Width (H):** 500 SPX Points
Fitting Methods: Anchor-Based, WLS, Robust Huber	**Option Structure:** 100-pt Fixed-Strike Butterflies

1. Executive Summary & Objective

The **SPX Multi-Date Volatility Surface Comparison Workspace** extends the existing fixed-strike butterfly dashboard into a multi-session cross-sectional analytics laboratory. Rather than analyzing a single date with synthetic shock sliders, this workspace imports any number of historical trading dates and solves three core quantitative objectives:

- **Objective 1 (Cross-Date Smile Measurement):** Quantify the evolution of the market's central volatility (Level), put-vs-call slope (Skew), and wing convex bend (Curvature) across dates, separating broad market factor shifts from localized strike-specific supply/demand anomalies (Residuals).
- **Objective 2 (5-Stage Transformation Bridge):** Construct an exact, stage-by-stage bridge showing how the Reference Date's smile transforms into the Focus Date's smile: IV_0 → +Level (IV_L) → +Skew (IV_LS) → +Curvature (IV_LSC) → Actual Observed Smile (IV_Focus), validating exact numerical closure.
- **Objective 3 (Fixed-Strike Fly Repricing vs. Rolling Observed Flies):** Rigorously distinguish between *observed changes in daily rolling ATM-relative butterfly quotes* (which reflect newly quoted contracts as spot drifts) and *controlled fixed-reference IV factor repricing* (which holds spot, strikes, and DTE constant while isolating pure factor sensitivities).

2. Mathematical Formulation & Factor Derivation

2.1 Coordinate System Normalization

Let K denote the option strike price, S_0 the underlying SPX spot index level, and K_{ATM} the at-the-money strike price for the ~60 DTE contract. The raw strike distance is defined as:

$$\text{Raw Offset:} \quad \Delta K = K - K_{ATM} \qquad | \qquad \text{Normalized Coordinate:} \quad x = \frac{(K - K_{ATM})}{H}$$

Why normalize by H? SPX spot fluctuates over time. A 500-point distance represents a standard wing boundary. Normalizing by reference width H (default 500 SPX points) creates a dimension-free coordinate x where:

- **x = 0:** Exactly at-the-money (K = K_ATM)
- **x = +1:** Right wing anchor (K = K_ATM + 500, Out-of-the-Money Call)
- **x = –1:** Left wing anchor (K = K_ATM – 500, Out-of-the-Money Put)

2.2 Single-Date Factor Definitions

For any single trading session, let σ(x) denote the implied volatility smile expressed in percentage points (e.g. 15.5% = 15.5). The smile is characterized by three canonical measurements:

Factor	Mathematical Definition	Financial Interpretation
Level	**Level = σ(0) = σ(K_ATM)**	The baseline market volatility anchor at the ATM strike.
Skew (S)	**Skew = σ(+1) − σ(−1)** **= σ(ATM+H) − σ(ATM−H)**	Slope across the wings. In equity index options, OTM puts trade higher than OTM calls, yielding S < 0.
Curvature (C)	**Curvature = [σ(−1) + σ(+1)] / 2 − σ(0)**	Convex bend of the smile. Measures how much the average wing volatility sits above the ATM anchor.

2.3 Mathematical Proof: Second-Order Polynomial Representation

Consider a parabolic representation of the smile around the ATM strike: **σ(x) ≈ a_0 + a_1 · x + a_2 · x²**. We now prove that setting a_0 = Level, a_1 = Skew / 2, and a_2 = Curvature reproduces the smile exactly at all three anchor points x ∈ {−1, 0, +1}:

1. At x = 0: σ(0) = a_0 = Level ⇒ a_0 = Level
2. At x = +1: σ(+1) = Level + a_1(1) + a_2(1)² = Level + a_1 + a_2
3. At x = −1: σ(−1) = Level + a_1(−1) + a_2(−1)² = Level − a_1 + a_2
4. Subtracting (2) − (3): σ(+1) − σ(−1) = 2 · a_1 ⇒ a_1 = Skew / 2
5. Adding (2) + (3): σ(+1) + σ(−1) = 2 · Level + 2 · a_2
⇒ a_2 = [σ(+1) + σ(−1)] / 2 − Level ⇒ a_2 = Curvature

Conclusion: The polynomial **σ(x) = Level + (Skew / 2) · x + Curvature · x²** is the unique quadratic passing through the three anchor points. The linear term coefficient is Skew / 2 because the span from x = −1 to x = +1 covers a distance of Δx = 2, making the average slope (σ(+1) − σ(−1)) / 2 = Skew / 2.

2.4 Two-Date Factor Differences (Reference Date 0 vs. Focus Date 1)

When comparing a Focus Date (Date 1) against a Reference Date (Date 0), the factor changes are defined as:

- Δ**Level (L):** L = Level_1 − Level_0 = σ_1(0) − σ_0(0)
- Δ**Skew (S):** S = Skew_1 − Skew_0 = [σ_1(+1) − σ_1(−1)] − [σ_0(+1) − σ_0(−1)]
- Δ**Curvature (C):** C = Curvature_1 − Curvature_0 = [Curv_1 − Curv_0]

The *modelled volatility shift* at any normalized offset x is: **Δσ_model(x) = L + (S / 2) · x + C · x²**
And the *local unexplained residual* is: **ε(x) = [σ_1(x) − σ_0(x)] − Δσ_model(x)**.

Exact Anchor Identity:
- *At x = 0:* Δσ_model(0) = L = σ_1(0) − σ_0(0) ⇒ ***ε(0) ≡ 0.0000***
- *At x = +1:* Δσ_model(+1) = L + S/2 + C = σ_1(+1) − σ_0(+1) ⇒ ***ε(+1) ≡ 0.0000***
- *At x = −1:* Δσ_model(−1) = L − S/2 + C = σ_1(−1) − σ_0(−1) ⇒ ***ε(−1) ≡ 0.0000***

Under Anchor-Based estimation, the model fits the three reference strikes with zero residual error. Any non-zero residual ε(x) at intermediate strikes (e.g. ±100, ±200, ±300, ±400) reflects genuine local smile deformation.

3. 5-Stage Smile Transformation Bridge

To make smile evolution transparent to traders and risk managers, the model breaks the transition from Reference Smile (IV_0) to Focus Smile (IV_Focus) into 5 cumulative transformation layers:

Stage	Mathematical Formula	Incremental Shift	Traded Meaning
1. IV_0	**IV_0(x) = σ_ref(x)**	Baseline	Reference date historical smile.
2. IV_L	**IV_L(x) = IV_0(x) + L**	+ L	Parallel market-wide volatility level shift.
3. IV_LS	**IV_LS(x) = IV_L(x) + (S/2)·x**	+ (S/2)·x	Linear tilt across puts and calls (Skew rotation).
4. IV_LSC	**IV_LSC(x) = IV_LS(x) + C·x²**	+ C·x²	Quadratic smile curvature (wing bending).
5. Actual	**IV_Actual(x) = IV_LSC(x) + ε(x)**	+ ε(x)	Actual focus smile, including local residual gaps.

Attribution Identity: For every aligned strike offset x, the sum of the factor contributions and local residual identically reconstructs the total observed volatility change:

$\Delta\sigma$_**observed(x) ≡ LevelComponent + SkewComponent(x) + CurvatureComponent(x) + Residual(x)**

This identity is verified in automated unit test suite test_multidate_comparison.py with error < 10■¹².

4. Fixed-Strike Butterfly Construction & Repricing

4.1 Butterfly Spread Structure

A standard butterfly spread with wing width *h = 100* points is defined as:

- **Long 1 Lower Wing:** Strike K_low = K_mid – 100
- **Short 2 Middle Body:** Strike K_mid
- **Long 1 Upper Wing:** Strike K_upp = K_mid + 100

The net market premium is: **F_mkt = P_mkt(K_low) – 2 · P_mkt(K_mid) + P_mkt(K_upp)**.

4.2 Critical Distinction: Observed Rolling Difference vs. Fixed Repricing

Traders often conflate two very different historical comparison concepts. The dashboard implements and labels both:

Comparison Framework	What It Calculates	Financial Meaning & Limitations
1. Observed Daily Differences	**ΔF_obs = F_1(ATM_1 + ΔK) – F_0(ATM_0 + ΔK)**	Compares *newly quoted daily ATM-relative butterflies*. Because the ATM strike moves with spot, this does **NOT** represent the holding-period P&L; of an existing position. It represents how pricing for equivalent-delta structures has shifted.
2. Fixed-Reference Factor Repricing	**Freezes Date 0 Spot (S_0), Strikes (K), and DTE (T_0). Reprices legs under L, S, C shocks.**	Controlled laboratory experiment. Measures how much of the butterfly's premium change was driven by market level shift vs. skew tilt vs. curvature bend vs. residual gap.

4.3 Sequential Fly Attribution Identity

Under fixed-reference repricing, each leg is repriced using the baseline-preserving Black-Scholes formula:

$P_{reprice} = P_{orig} + BS(S_0, K, T_0, \sigma_{shocked}) - BS(S_0, K, T_0, \sigma_{orig})$

This preserves the market quote basis while capturing exact non-linear sensitivity to volatility. Sequential butterfly stages are computed as:

- **F_0:** Original Reference Date fly premium
- **F_L:** Repriced under Level shift (L)
- **F_{LS}:** Repriced under Level + Skew shift $(L + (S/2)\cdot x)$
- **F_{LSC}:** Repriced under 3-Factor model $(L + (S/2)\cdot x + C\cdot x^2)$
- **F_{Full}:** Repriced under Full observed smile shift (including local residual)

Sequential effects are then attributed as:

- **Level Effect = $F_L - F_0$**
- **Skew Effect = $F_{LS} - F_L$**
- **Curvature Effect = $F_{LSC} - F_{LS}$**
- **Residual Effect = $F_{Full} - F_{LSC}$**

Exact Identity: Total IV-Only Change ($F_{Full} - F_0$) ≡ Level + Skew + Curvature + Residual Effect

5. Strict Market-Only Quote Safeguard

The Excel Data Structure: The historical dataset (sticky_delta_2026-10-12_2025-06-05.xlsx) records SPX option chains where strike offset 0 contains an ATM Call quote, but *no separate ATM Put quote* is recorded. An out-of-the-money put butterfly centered at ATM – 100 has legs:

- Lower Wing: Strike = ATM – 200 (Put quote available)
- Middle Body: Strike = ATM – 100 (Put quote available)
- Upper Wing: Strike = ATM (Requires Put quote, which is missing from source data)

Why Synthetic Put Parity is Dangerous: Inferring the ATM put via put-call parity $P = C - S + K$ requires dividend and borrowing cost assumptions that introduce synthetic noise into an otherwise pure market dataset.

The Strict Market Safeguard: The model omits the incomplete ATM – 100 Put Butterfly in strict mode. Complete put butterflies from ATM – 200 down to ATM – 800 are computed strictly from real market quotes. A permanent UI notification clearly alerts the user of this safeguard.

6. Numerical Benchmark Walkthrough: Aug 3 vs. Aug 4, 2026

To allow complete external verification, the table below provides the exact audited numbers for the canonical benchmark dates (Reference: August 3, 2026 vs. Focus: August 4, 2026) with reference width H = 500 SPX points:

Metric / Parameter	August 3, 2026 (Ref)	August 4, 2026 (Focus)	Difference (Focus – Ref)
SPX Underlying Spot (S_0)	6,380.00	6,420.00	+40.00 pts
ATM Strike (K_{ATM})	7,230.00	7,375.00	+145.00 pts
Target DTE (Maturity)	60 Days	60 Days	0 Days
ATM IV Level (x = 0)	15.4210%	16.0292%	**ΔLevel = +0.6082 pts**
Smile Skew (S at H=500)	–2.8410 pts	–1.7578 pts	**ΔSkew = +1.0832 pts**
Smile Curvature (C at H=500)	+1.4820 pts	+1.4618 pts	**ΔCurvature = –0.0202 pts**
Call Fly @ +500 (Prem)	12.1470 SPX pts	10.8533 SPX pts	**Observed Δ = –1.2937 pts**

7. Codebase Architecture & Implementation Guide

The multi-date enhancement is implemented cleanly in dedicated modules to ensure 100% backward compatibility with the existing single-date lab. Here is the architectural mapping:

File Path	Core Responsibility	Key Functions & Classes
spx_butterfly/multidate_model.py	Pure mathematical calculations, factor fits, smile decomposition, fly repricing.	calculate_daily_factors() fit_factors_between_dates() decompose_iv_smile() compute_fixed_fly_attribution()
spx_butterfly/multidate_charts.py	Plotly figure generators with harmonized qualitative date colors and hover spikes.	create_multidate_iv_chart() create_iv_attribution_chart() create_multidate_fly_chart() create_historical_factor_trends_chart()
spx_butterfly/multidate_view.py	Streamlit UI layout for the multi-date comparison workspace (Rows 1–8).	render_multidate_workspace() Date Presets & Multiselect sync Cross-Sectional Strike Matrix Leg-Level Single Fly Inspector
spx_butterfly/single_date_view.py	Completely preserved original single-date butterfly scenario lab.	render_single_date_workspace()
app.py	Application entry point with dual-workspace top & sidebar navigation.	Workspace routing & data loading cache.

8. How to Verify, Audit, and Improve the System

Checklist for Verification:

1. **Run Automated Tests:** Execute pytest -v in the terminal. All 49 tests should pass in under 12 seconds.
2. **Verify Identity 1 (Smile Reconstruction):** In spx_butterfly/multidate_model.py, check that obs_change = level_comp + skew_comp + curv_comp + residual holds to < 10■¹² at all strikes.
3. **Verify Identity 2 (Fly Attribution Closure):** Verify that total_iv_change = level_eff + skew_eff + curv_eff + residual_eff.
4. **Inspect the UI Matrix:** Open http://localhost:8501, expand the *'Cross-Sectional Strike-by-Strike Comparative Matrix'*, and check that the Δ (Focus – Ref) column exactly matches the difference between the two dates' IV columns.

Recommended Pathways for Improvement:

• **Higher-Order Smile Terms:** Add a quartic term (C_4 · x■) to capture 'wing curvature' (excess kurtosis) independently from ATM curvature.
• **Term Structure Dimension:** Extend the 60 DTE comparison to multi-tenor surfaces (30, 60, 90 DTE) to evaluate term-structure shifts.
• **Natural Cubic Spline Interpolation:** Replace linear node interpolation with tensioned cubic splines or SVI (Stochastic Volatility Inspired) parametric curves.